

Parity Violation in Gravity: New Constructions and Inflationary Signals

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Modified Gravity Workshop - ICASU 2023

Based on 2207.05094 w/ Stephon Alexander
and 2303.04815 with Stephon, Marc Kamionkowski, & Oliver Philcox

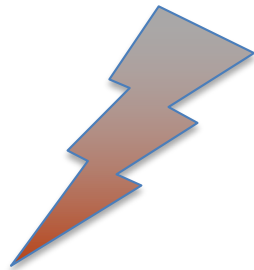
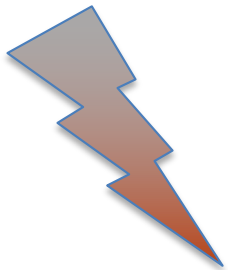
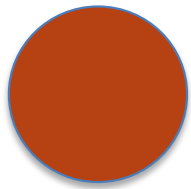
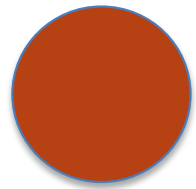
Roadmap

I will...

- I) Display two new constructions (i.e types of UV completions) of dynamical Chern-Simons (dCS) gravity that encompass a wide range of EFT scales
- II) Show that dCS gravity yields a parity-violating primordial scalar trispectrum that can be made large.
- III) Demonstrate that ratios of odd and even trispectra contain direct information about spin.
- IV) Quantify how well upcoming and future experiments can constrain scalar trispectra.
- V) Hint to how parity-violating trispectra can constrain baryogenesis.

What is $\mathbb{P}$ arity? How can Gravity violate it?

$$\mathbb{P}\mathbf{x} = -\mathbf{x}$$



$$A_L - A_R$$

Chern-Simons Gravity

What is [dynamical] Chern-Simons (dCS) Gravity?

$$S = \int d^4x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} (\partial_\mu \phi) (\partial_\nu \phi) + \frac{M_{\text{pl}}^2}{2} R - \frac{\phi}{4f} *RR \right]$$

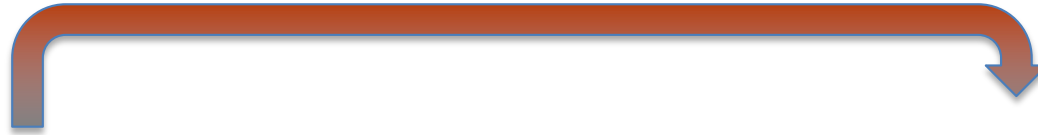
Spacetime Metric, Lorentzian Signature
 dCS pseudo-scalar
 Einstein-Hilbert Term
 dCS Decay Constant
 Pontryagin Density

Effective Theory

$*RR = *R^\rho{}_\sigma{}^{\mu\nu} R^\sigma{}_{\rho\mu\nu}$
 $*R^\rho{}_\sigma{}^{\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\alpha\beta} R^\rho{}_{\sigma\alpha\beta}$

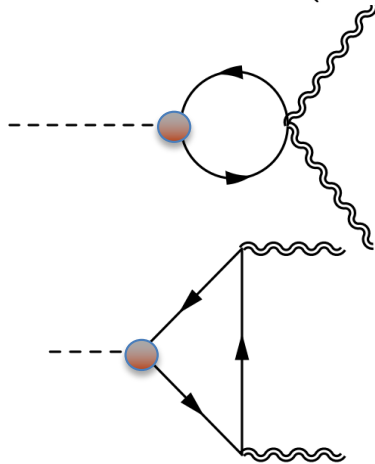
New Construction I: Spontaneous Symmetry Breaking

$$\Phi = (1/\sqrt{2})(F + \sigma) \exp[ia(x)/F]$$



$$\mathcal{L}_\Phi = \sum_{j=L,R} i\bar{\Psi}_j \gamma^\mu \partial_\mu \Psi_j - \tilde{m}_\Psi (\bar{\Psi}_R \Psi_L + \bar{\Psi}_L \Psi_R) \\ + \partial_\mu \Phi \partial^\mu \Phi^* - V(|\Phi|^2) - (y_\Phi \Phi \bar{\Psi}_L \Psi_R + \text{h.c.})$$

$$\mathcal{L}_\Phi = \bar{\Psi} \left[i\gamma^\mu \partial_\mu - \left(\tilde{m}_\Psi + \frac{y_\Phi F}{\sqrt{2}} \right) \right] \Psi \\ + \frac{1}{2} \partial_\mu a \partial^\mu a - \frac{y_\Phi}{\sqrt{2}} a \bar{\Psi} \gamma^5 \Psi,$$



$$\mathcal{L}_g^{\text{eff}} = -\frac{g}{384\pi^2} \frac{a}{2m_\Psi} * RR$$

$$f = 192\sqrt{2}\pi^2 \frac{m_\Psi}{y_\Phi}$$

New Construction 2: Dynamical Symmetry Breaking

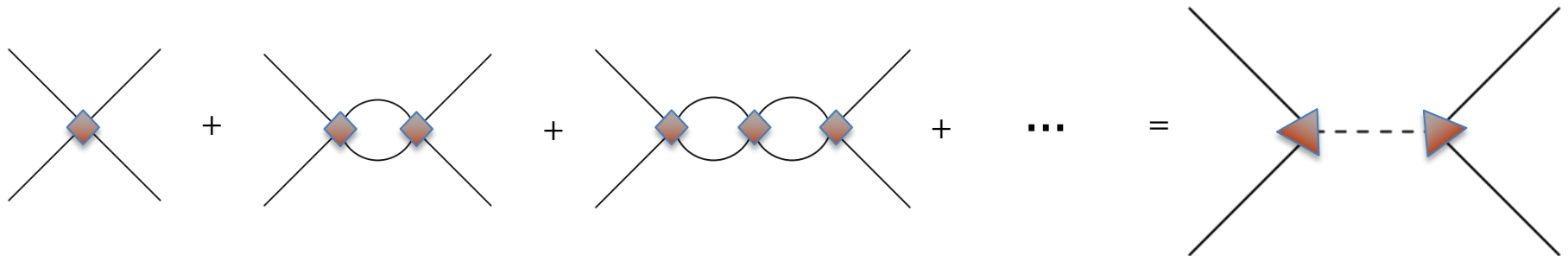
$$\mathcal{L}_\Psi = \bar{\Psi} (i\gamma^\mu \partial_\mu - \tilde{m}_\Psi) \Psi - \lambda \bar{\Psi} \Psi \bar{\Psi} \Psi$$

★ Cutoff Λ

“bare” mass

Dirac
Fermion

(Attractive) Fermion
Self-Coupling Constant



New Construction 2: Dynamical Symmetry Breaking

$$Z_\alpha = \int \mathcal{D}\alpha \mathcal{D}\bar{\alpha} \exp \left(- \int d^4x \tilde{m}_\Phi^2 \bar{\alpha} \alpha \right)$$

+

$$\Psi = \Psi_\ell + \Psi_s$$

$$\Phi = \alpha - \tilde{m}_\Phi^{-2} \bar{\Psi}_s \Psi_s$$

+

$\Rightarrow$

$$\begin{aligned} \tilde{\mathcal{L}}_\Psi = & \bar{\Psi} (i\gamma^\mu \partial_\mu - \tilde{m}_\Psi) \Psi - \lambda \bar{\Psi} \Psi \bar{\Psi} \Psi \\ & + (\partial_\mu \Phi^*) (\partial^\mu \Phi) - \underline{y_\Phi (\Phi \bar{\Psi} \Psi + \text{h.c.})} \\ & + m_\Phi^2 |\Phi|^2 - \frac{\lambda_\Phi}{4} |\Phi|^4, \end{aligned}$$



Integrate Out Short – Scale Modes

$$y_\Phi \sim \lambda \Lambda^2$$

$$f = 192 \sqrt{2} \pi^2 \frac{m_\Psi}{y_\Phi}$$



arXiv: 2207.05094, Collaborator: Stephon Alexander

What are Observable Signatures of dCS?

(see Leah Jenks' talk tomorrow)

- 1) Compact Binary Coalescence (CBC) Waveform-Modification
 - 2) Cosmic Birefringence of CBC Gravitational Waves
 - 3) Modification of Orbits around Earth
- } Late-Time
- $f \ll \text{eV}$

- 4) Polarized Primordial Gravitational Waves
 - 5) Primordial Imprints on Galaxy Clustering and the CMB
- } Early-Time

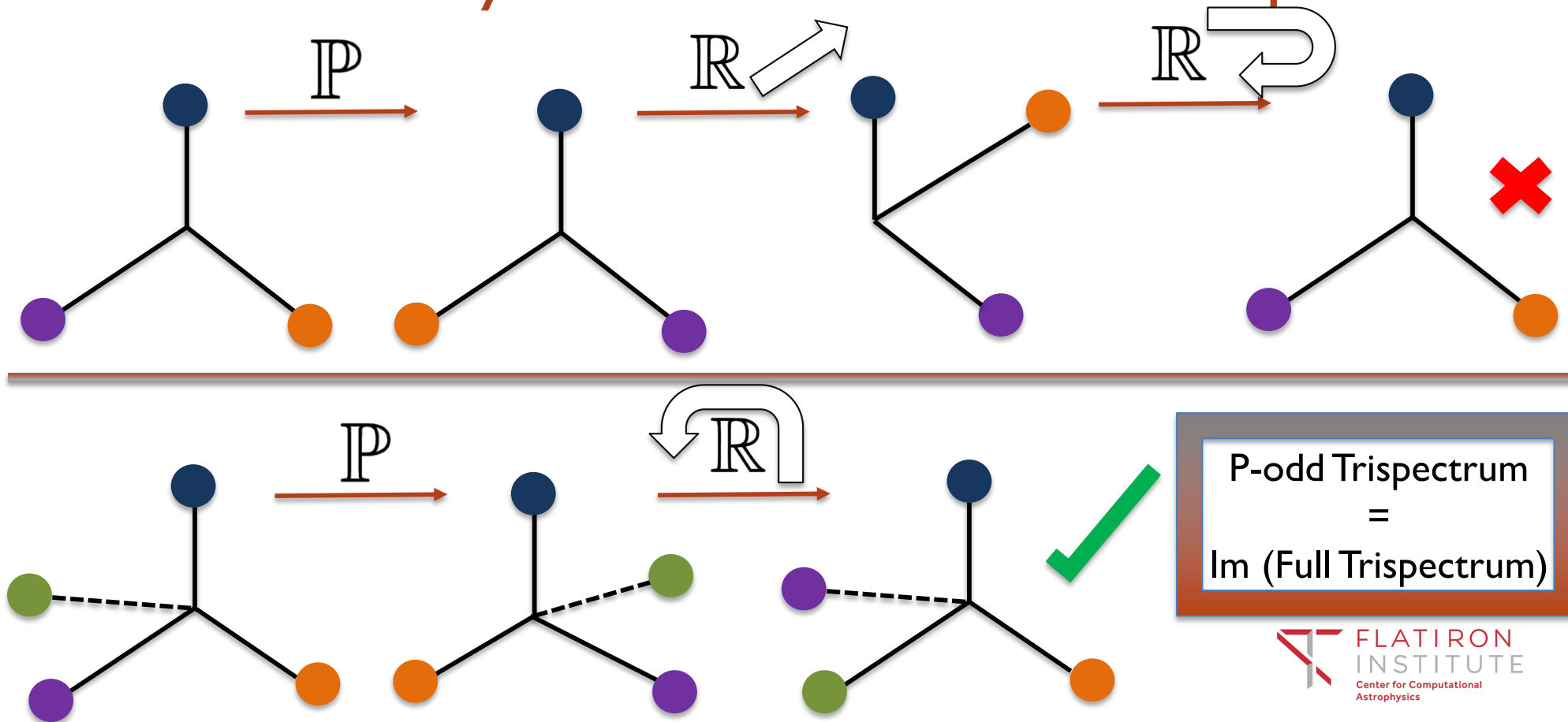


How is Galaxy Clustering Described?

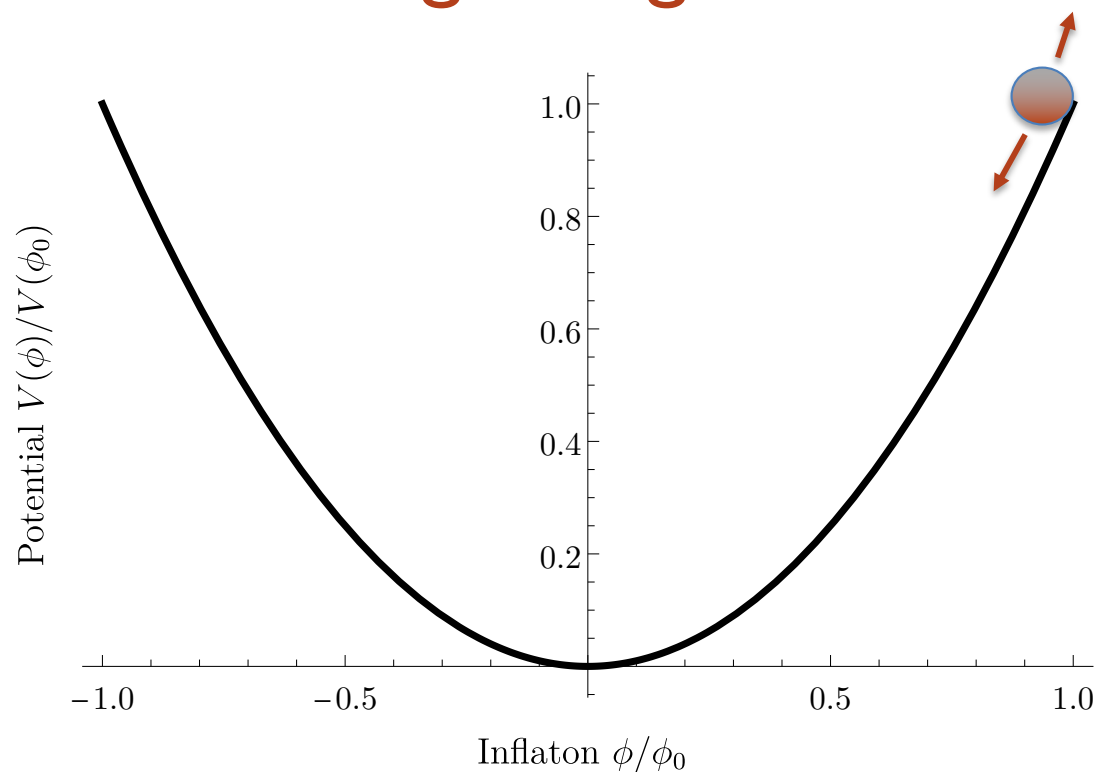
Galaxy N-Point Functions!

$$\delta_g(\mathbf{x}) = \frac{n_g(\mathbf{x}) - \bar{n}_g}{\bar{n}_g}$$

Enter: The Galaxy Four Point Function / Trispectrum



Inflation: A Lightning Review



Slow-Roll Parameter

$$\varepsilon = \frac{1}{2} \frac{\dot{\phi}^2}{M_{\text{pl}}^2 H^2}$$

Solves:

- 1) The Horizon Problem
- 2) The Flatness Problem
- 3) Magnetic Monopole Problem

Quantum Inflaton Fluctuations Seed
Density Perturbations

dCS Inflation: Full Action

$$S = \int d^4x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} (\partial_\mu \phi) (\partial_\nu \phi) + \frac{M_{\text{pl}}^2}{2} R - \frac{\phi}{4f} *RR \right]$$

Spacetime Metric, Lorentzian Signature $\uparrow$
 dCS pseudoscalar/Inflaton $\uparrow$
 Einstein-Hilbert Term $\uparrow$
 Pontryagin Density $\uparrow$
 dCS Decay Constant $\uparrow$

$$g_{\mu\nu} = a^2(\tau) \begin{pmatrix} -1 & 0 \\ 0 & \delta_{ij} + h_{ij} \end{pmatrix}$$

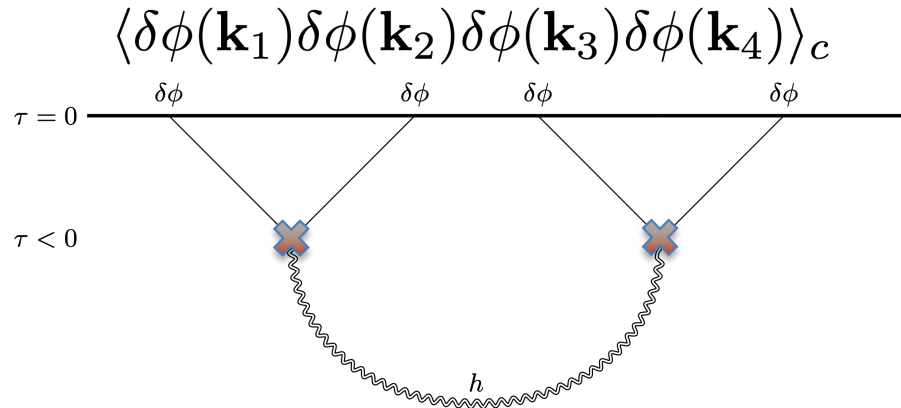
$$*RR = *R^\rho{}_\sigma{}^{\mu\nu} R^\sigma{}_{\alpha\mu\nu}$$

$$*R^\rho{}_\sigma{}^{\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\alpha\beta} R^\rho{}_{\sigma\alpha\beta}$$

Inflaton Potential: $V \sim \Lambda^4$

$$\Lambda \sim \sqrt{H M_{\text{pl}}} = 10^{16} \text{ GeV} [H/(10^{14} \text{ GeV})]^{1/2}$$

Primordial Scalar Trispectrum from dCS



$$\propto P_{\delta\phi} P_{\delta\phi} P_h$$

$$\zeta(\mathbf{k}) = -H[\delta\phi(\mathbf{k})/\dot{\phi}]$$

Perturbed Inflaton Kinetic Term

$$\frac{1}{2}\sqrt{-g}g^{\mu\nu}\partial_\mu\delta\phi\partial_\nu\delta\phi = \frac{1}{2}a^2\eta^{\mu\nu}\partial_\mu\delta\phi\partial_\nu\delta\phi + \boxed{\frac{1}{2}a^2h^{ij}\partial_i\delta\phi\partial_j\delta\phi}$$

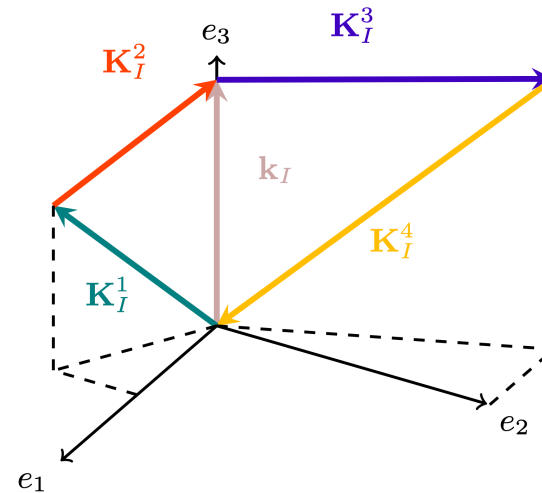
$$\mathbf{K}_s = \{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4\}$$

$$\mathbf{K}_t = \{\mathbf{k}_1, \mathbf{k}_3, \mathbf{k}_2, \mathbf{k}_4\}$$

$$\mathbf{K}_u = \{\mathbf{k}_1, \mathbf{k}_4, \mathbf{k}_3, \mathbf{k}_2\}$$

$$\mathbf{K}_I^1 + \mathbf{K}_I^2 = -\mathbf{K}_I^3 - \mathbf{K}_I^4 = \mathbf{k}_I$$

$$I \in \{s, t, u\}$$



$$\begin{aligned}\hat{\mathbf{e}}_1 &= R_I \hat{\mathbf{x}} \\ \hat{\mathbf{e}}_2 &= R_I \hat{\mathbf{y}} \\ \hat{\mathbf{e}}_3 &= R_I \hat{\mathbf{z}}\end{aligned}$$

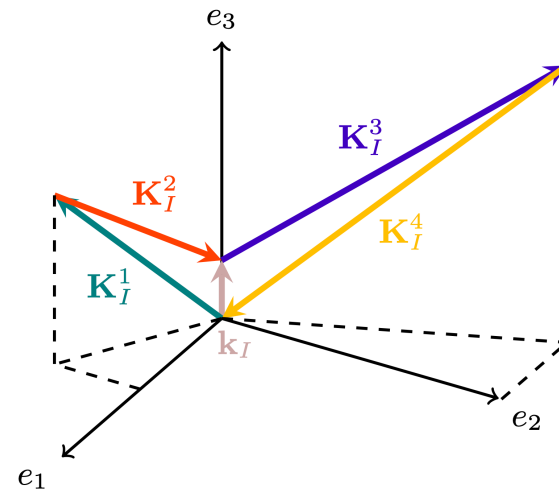
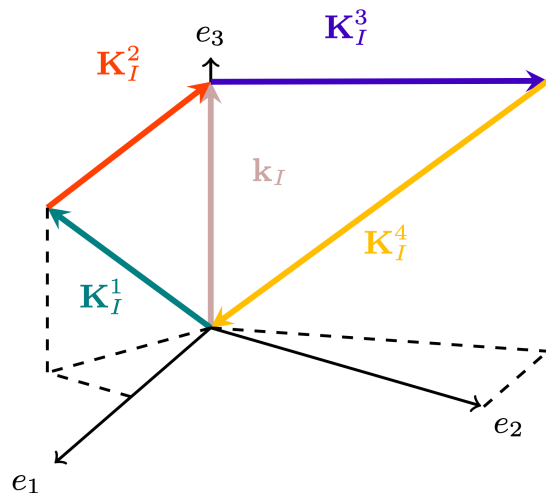
$$\hat{\mathbf{e}}_3 = \hat{\mathbf{k}}_I$$

FLATIRON
INSTITUTE
Center for Computational
Astrophysics

The Collapsed Limit

$$\mathbf{k}_i \gg \mathbf{k}_I \approx 0, \quad i \in \{1, 2, 3, 4\}$$

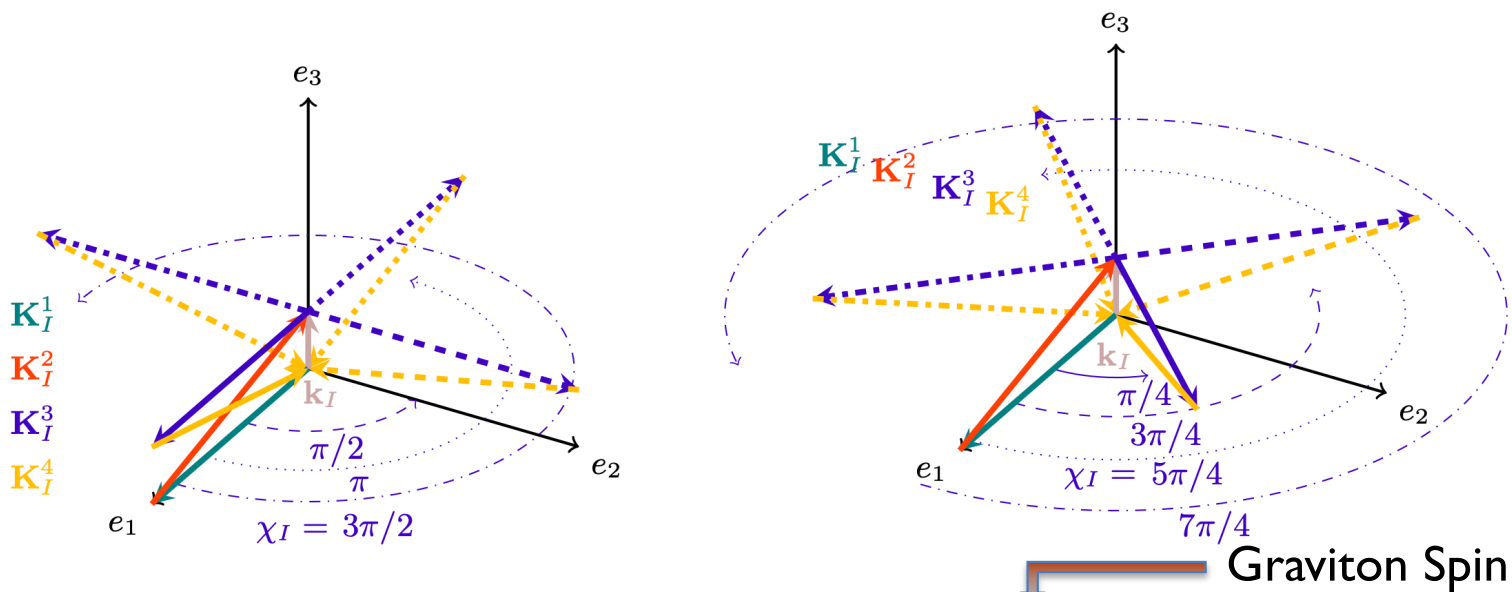
$$K_I^1 \approx K_I^2 \quad | \quad K_I^3 \approx K_I^4$$



The Collapsed Limit Scalar Trispectra

$$\text{Re} \langle \zeta(\mathbf{k}_1) \zeta(\mathbf{k}_1) \zeta(\mathbf{k}_3) \zeta(\mathbf{k}_4) \rangle_c'^{\text{GE}} \Big|_{I-\text{Coll.}} = \frac{9}{16} r \cos(2\chi_I) \sin^2(\theta_I^1) \sin^2(\theta_I^3) P_\zeta(k_I) P_\zeta(K_I^1) P_\zeta(K_I^3) \underline{\underline{\times F}}$$

$$\text{Im} \langle \zeta(\mathbf{k}_1) \zeta(\mathbf{k}_1) \zeta(\mathbf{k}_3) \zeta(\mathbf{k}_4) \rangle_c'^{\text{GE}} \Big|_{I-\text{Coll.}} = \frac{9}{16} \Pi_{\text{circ}} r \sin(2\chi_I) \sin^2(\theta_I^1) \sin^2(\theta_I^3) P_\zeta(k_I) P_\zeta(K_I^1) P_\zeta(K_I^3) \quad \underline{\times F}$$



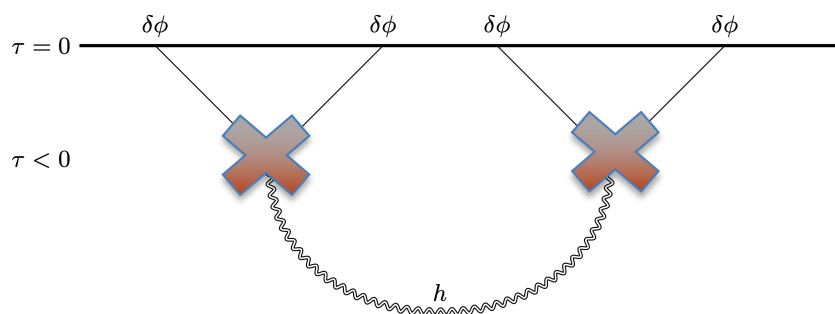
$$\text{Odd/Even} = \Pi_{\text{circ}} \cot(2\chi_I)$$



Two Examples Beyond Vanilla dCS Inflation

Superluminal Scalar Sound Speed

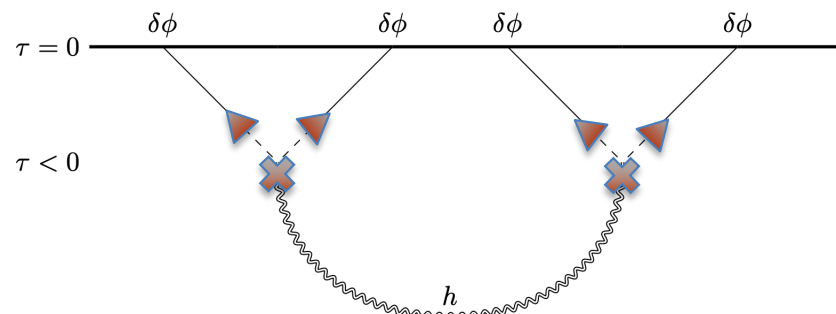
$$c_s^2 g^{ij} (\partial_i \phi) (\partial_j \phi) \quad \times$$



$$F = 2 \times 10^6 (c_s/6)^8$$

Quasi-Single Field/Multi-Field

$$a^3 \lambda \delta\sigma \dot{\delta\phi} \quad \triangle \quad h^{ij} \partial_i (\delta\sigma) \partial_j (\delta\sigma) \quad \times$$



$$F \sim 10^4 [w(\nu)/20]$$

$$\nu = \sqrt{9/4 - (m/H)^2} \lesssim 0.5$$

arXiv: 1504.05993
(Dimastrogiovanni, Fasiello, Kamionkowski)

$$F \sim 10^7 (N/60)^4$$

$$\nu \sim 3/2$$

arXiv: 0911.3380
(Xingang Chen, Yi Wang)



The Trispectra can be Large

arXiv: 2303.04815, Collaborators: Oliver Philcox, Stephon Alexander, Marc Kamionkowski

How Sensitive are we to Collapsed Trispectra?

$$[1 + \delta_{\text{odd}}^P (\Pi_{\text{circ}} - 1)] r \gtrsim 0.04 \left(\frac{n}{3}\right) \left(\frac{8 \times 10^5}{F}\right) \left(\frac{10^6}{N_{\text{modes}}}\right)^{1/2} \left(\frac{k_{\text{min}}}{0.003 \text{ h/Mpc}}\right)^{3/2} \left(\frac{0.3 \text{ h/Mpc}}{k_{\text{max}}}\right)^{3/2}$$

$$f \lesssim 4 \times 10^9 \text{ GeV} \left(\frac{3}{n}\right) \left(\frac{F}{8 \times 10^5}\right) \left(\frac{\varepsilon}{10^{-2}}\right)^{3/2} \left(\frac{H}{10^{14} \text{ GeV}}\right)^2 \left(\frac{N_{\text{modes}}}{10^6}\right)^{1/2} \left(\frac{0.003 \text{ h/Mpc}}{k_{\text{min}}}\right)^{3/2} \left(\frac{k_{\text{max}}}{0.3 \text{ h/Mpc}}\right)^{3/2}$$

$$N_{\text{modes}} = V \int_{k_{\text{min}}}^{k_{\text{max}}} \frac{d^3 \mathbf{k}}{(2\pi)^3} W(\mathbf{k}) \frac{[G^2(\mathbf{k}, \bar{z}) P_L^{gg}(\mathbf{k}, \bar{z})]^2}{[P_{\text{NL}}^{gg}(\mathbf{k}, \bar{z}) + \bar{n}^{-1}]^2}$$

What about Experiments?

$$[1 + \delta_{\text{odd}}^P (\Pi_{\text{circ}} - 1)] r \gtrsim 0.04 \left(\frac{n}{3}\right) \left(\frac{8 \times 10^5}{F}\right) \left(\frac{10^6}{N_{\text{modes}}}\right)^{1/2} \left(\frac{k_{\text{min}}}{0.003 h/\text{Mpc}}\right)^{3/2} \left(\frac{0.3 h/\text{Mpc}}{k_{\text{max}}}\right)^{3/2}$$

Experiments									
Name	$[z_{\text{min}}, z_{\text{max}}]$	f_{sky}	$[k_{\text{min}}, k_{\text{max}}] [h/\text{Mpc}]$	$\bar{n} [(h/\text{Mpc})^3]$	$\bar{b}$	$V [(\text{Gpc}/h)^3]$	N_{modes}	F	
BOSS ^a [112, 113]	[0.43, 0.7]	0.23	[0.01, 0.24]	3×10^{-4}	2.0	3.6	10^5	2×10^7	
DESI ^a [115]	[0.6, 1.7]	0.34	[0.003, 0.31]	3.8×10^{-4}	1.4	45	8×10^5	8×10^5	
Euclid ^a [114]	[0.9, 1.8]	0.36	[0.003, 0.34]	4.3×10^{-4}	1.7	44	10^6	6×10^5	
MegaMapper ^a [116, 117]	[2, 5]	0.34	[0.003, 0.64]	2.5×10^{-4}	3.8	155	7×10^6	10^5	
MSE ^a [120]	[1.6, 4]	0.24	[0.003, 0.54]	2.8×10^{-4}	3.7	91	6×10^6	10^5	
SPHEREx ^a [118, 119]	[0.1, 4.3]	0.65	[0.003, 0.46]	2.0×10^{-3}	1.1	360	2×10^7	10^5	
HIRAX ^b [121]	[0.8, 2.5]	0.36	[(0.01, 0.1), 0.38]	10^{-3}	1.9	88	3×10^6	$(2 \times 10^6, 7 \times 10^7)$	
PUMA-32K ^b [122]	[2, 6]	0.5	[(0.01, 0.1), 0.71]	7.6×10^{-3}	6.3	290	7×10^7	$(2 \times 10^5, 5 \times 10^6)$	

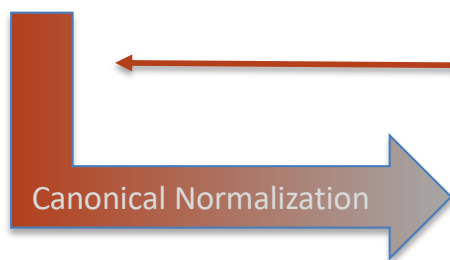
^aSpectroscopic Galaxy Survey

^b21-cm Experiment

dCS Inflation: Lepto/Baryogenesis

$$a^4 \frac{\phi(\tau)}{4f} * RR$$

$$\frac{\dot{\phi}}{4f} = \frac{1}{2} \frac{M_{\text{pl}}}{H} \left[\sqrt{\frac{\varepsilon}{2}} \frac{H^2}{M_{\text{pl}} f} \right] = \frac{1}{2} \frac{M_{\text{pl}}^2}{H} \underline{\mu}$$



Canonical Normalization

$$\partial_\mu J_5^\mu = *RR / (384\pi^2) \quad J_5^0 = n_L - n_R$$

$$\sim \underbrace{H\mu(n_L - n_R)}_{\text{Leptogenesis}}$$

Conclusions/Future Extensions

- 1) Two new constructions of dCS, yielding a large range of decay constants
- 2) First example of massless spin-exchange yielding a parity-violating scalar trispectrum
- 3) Scalar trispectra can probe dCS decay constants at very high values
- 4) Odd to even ratios, in collapsed limit, (i.e. the phase of the trispectrum)
give direct information about spin
- 5) Potential new probes of baryogenesis

dCS Extensions

- 1) Removal of the dCS Ghost
- 2) Parity-Odd Signal for all configurations
- 3) Where do the trispectra peak?
- 4) CMB signal

Baryogenesis Extensions

- 1) What type of spin-1 baryogenesis can be probed? (What vertex interactions, shapes, sensitivities, UV completions etc)
- 2) Other CMB/LSS probes?



Extra Content

The Gap Equation



$$\frac{\Delta m_\Psi}{m_\Psi} \left(\frac{2\pi^2}{\lambda\Lambda^2} \right) = 1 - \frac{m_\Psi^2}{\Lambda^2} \log \left(1 + \frac{\Lambda^2}{m_\Psi^2} \right), \quad \Delta m_\Psi = y_\Phi F / \sqrt{2}$$

$\Lambda \gg m_\Psi$

$$F \approx 0.9m_\Psi$$

$\Lambda \ll m_\Psi$

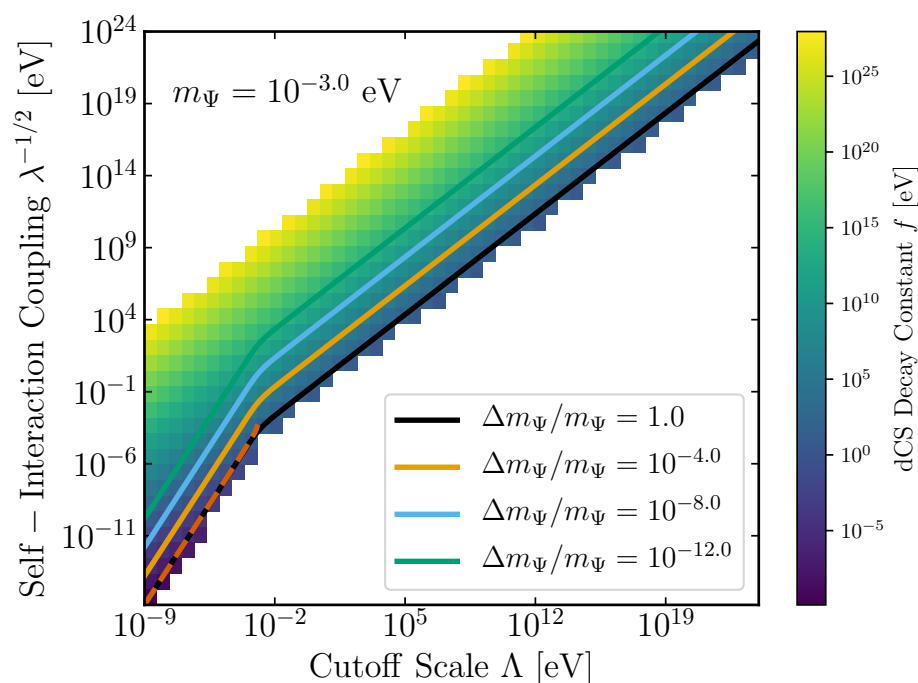
$$F \approx 0.45\Lambda^2/m_\Psi$$

Remember!



$$y_\Phi \sim \lambda\Lambda^2$$

General Parameter Space



$$\Lambda \gg m_\Psi$$

$$f = 1.7 \text{ eV} \left(\frac{\Delta m_\Psi}{m_\Psi} \right)^{-1} \left(\frac{m_\Psi}{10^{-3} \text{ eV}} \right)$$

$$\Lambda \ll m_\Psi$$

$$f = 0.85 \text{ eV} \left(\frac{\Delta m_\Psi}{m_\Psi} \right)^{-1} \left(\frac{\Lambda}{m_\Psi} \right) \left(\frac{\Lambda}{10^{-3} \text{ eV}} \right)$$

UV Completions for Fermion SI

(Scalar) Yukawa

$$\mathcal{L} \supset g_\chi \chi \bar{\Psi} \Psi$$

$$\Lambda \sim m_\chi$$

$$\lambda = (g_\chi / m_\chi)^2$$

$$f = 34 \text{ eV } g_\chi^{-2} \left(\frac{m_\Psi}{10^{-3} \text{ eV}} \right)$$

Gravitational Torsion

$$\mathcal{L} \supset (2/\gamma) R^{\mu\nu} \wedge e_\mu \wedge e_\nu$$

$$\Lambda \sim M_{\text{pl}}$$

$$\lambda = 3\pi / \Lambda_T^2$$

dCS Inflation: Quadratic Action

$$S^{(2)} = \int d^4x \left\{ \underbrace{-\frac{1}{2}a^2 (\delta\phi')^2 - \frac{1}{2}a^2 (\partial_i \delta\phi) (\partial^i \delta\phi)}_{\text{Inflaton Kinetic Term}} + \frac{M_{\text{Pl}}^2}{8} a^2 \underbrace{\left[(h_j^i)' (h_i^j)' + (\partial_k h_j^i) (\partial^k h_i^j) \right]}_{\text{Einstein-Hilbert Term}} - a^4 \frac{\phi(\tau)}{4f} {}^*RR \right\}$$

★ Canonical Normalization Required

Background Inflaton

↑

Pontryagin Density

↑

dCS Inflation: Dynamics and Primordial Power Spectra

$$u''(\tau, \mathbf{k}) - \frac{2}{\tau} u'(\tau, \mathbf{k}) + k^2 u(\tau, \mathbf{k}) = 0,$$

$$u''_{\pm}(\tau, \mathbf{k}) - \frac{2}{\tau} u'_{\pm}(\tau, \mathbf{k}) + \left(k^2 \mp \frac{2k\mu}{\tau} \right) u_{\pm}(\tau, \mathbf{k}) = 0,$$

$$u(\tau, \mathbf{k}) = \frac{H}{\sqrt{2k^3}} (1 + ik\tau) e^{-ik\tau}$$

$$u_{\pm}(\tau, \mathbf{k}) = \frac{H}{M_{\text{pl}} \sqrt{k^3}} (1 + ik\tau) e^{-ik\tau} \left[\frac{U(2 \mp i\mu, 4, 2ik\tau)}{U(2, 4, 2ik\tau)} \right] \exp\left(\pm \frac{\pi}{2} \mu\right)$$

$$\Pi_{\text{circ}} = 0.9 \left(\frac{\varepsilon}{10^{-2}} \right)^{1/2} \left(\frac{H}{10^{14} \text{ GeV}} \right)^2 \left(\frac{10^9 \text{ GeV}}{f} \right)$$



Collaborators: Oliver Philcox, Stephon Alexander, Marc Kamionkowski

dCS Inflation: Dynamics and Primordial Power Spectra

$$P_{\zeta}(k) = \frac{2\pi^2}{k^3} A, \quad A = 2.1 \times 10^{-9}$$

$$P_t(k) = \frac{2\pi^2}{k^3} r A, \quad r = \Theta_{\text{even}} \bar{r}$$

$$\Theta_{\text{even}} = 1 + \frac{2\pi^2 - 3}{3} \mu^2, \quad \bar{r} = 16\varepsilon$$

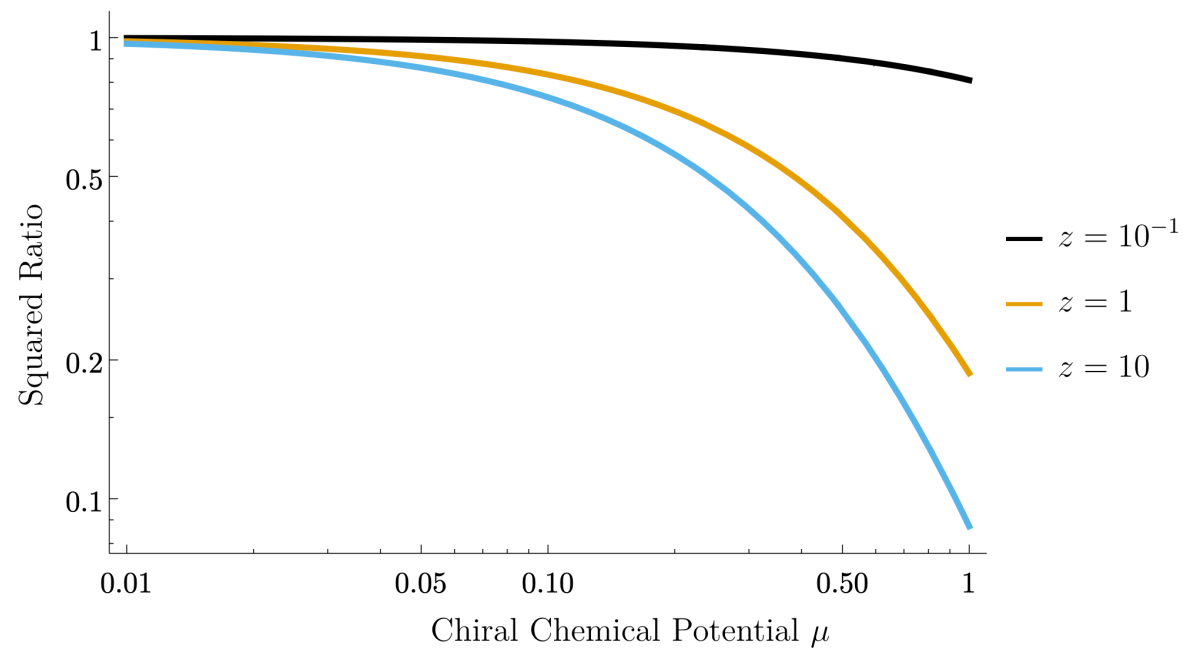
Full Graviton-Exchange Trispectrum Shape

$$\begin{aligned}
 & -\frac{\bar{A}_t}{k_I^3} \frac{H^4}{\left[\prod_{i=1}^4 2k_i^3\right]} \left\{ \frac{K_I^1 + K_I^2}{[a_{34}^I]^2} \left[\frac{1}{2} (a_{34}^I + k_I) \left([a_{34}^I]^2 - 2b_{34}^I \right) + k_I^2 (K_I^3 + K_I^4) \right] + [(1, 2) \leftrightarrow (3, 4)] \right. \\
 & \quad + \frac{K_I^1 K_I^2}{k_t} \left[\frac{b_{34}^I}{a_{34}^I} - k_I + \frac{k_I}{a_{12}^I} \left(K_I^3 K_I^4 - k_I \frac{b_{34}^I}{a_{34}^I} \right) \left(\frac{1}{k_t} + \frac{1}{a_{12}^I} \right) \right] + [(1, 2) \leftrightarrow (3, 4)] \\
 & \quad \left. - \frac{k_I}{a_{12}^I a_{34}^I k_t} \left[b_{12}^I b_{34}^I + 2k_I^2 \left(\prod_{i=1}^4 k_i \right) \left(\frac{1}{k_t^2} + \frac{1}{a_{12}^I a_{34}^I} + \frac{k_I}{k_t a_{12}^I a_{34}^I} \right) \right] \right\}
 \end{aligned}$$

$$a_{ij}^I = K_I^i + K_I^j + k_I$$

$$b_{ij}^I = (K_I^i + K_I^j)k_I + K_I^i K_I^j \quad k_t = \sum_{i=1}^4 k_i$$

Amplitude-Approximation Validity Plot



Non-Linearities, Foregrounds, and Noise, Oh My

$$N_{\text{modes}} = V \int_{k_{\min}}^{k_{\max}} \frac{d^3 \mathbf{k}}{(2\pi)^3} W(\mathbf{k}) \frac{[G^2(\mathbf{k}, \bar{z}) P_L^{gg}(\mathbf{k}, \bar{z})]^2}{[P_{\text{NL}}^{gg}(\mathbf{k}, \bar{z}) + \bar{n}^{-1}]^2}$$

$$P_a^{gg}(\mathbf{k}, z) = [b(z) + f(z)\mu^2]^2 D^2(z) P_a^m(k)$$

$$G(\mathbf{k}, z) \simeq \exp \left[-\frac{1}{2} \left(k_{\perp}^2 + k_{\parallel}^2 [1 + f(z)]^2 \right) \Sigma^2(z) \right] \quad \Sigma(z) = \int_0^{\infty} \frac{dk}{6\pi^2} P_L^m(k, z)$$

$$W(\mathbf{k}) = \Theta \left\{ k_{\parallel} - \max \left[k_{\perp} \frac{\chi(z)H(z)}{(1+z)} \sin[\theta_w(z)], k_{\parallel \min} \right] \right\}, \quad \theta_w(z) = N_w \frac{1.22}{2\sqrt{0.7}} \frac{\lambda_{\text{obs}}^{21}(z)}{D_{\text{phys}}}$$

Angular Integrals!

$$\begin{aligned} \left(\frac{S}{N} \right)_{rF}^2 &= \frac{(rF)^2}{24} \int_{k_{\min}}^{k_{\max}} \left[\prod_{i=1}^4 \frac{d^3 \mathbf{k}_i}{(2\pi)^3} \right] \frac{[S_{\text{GE}}^{\text{even}}(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4)]^2}{P(k_1)P(k_2)P(k_3)P(k_4)} \\ &= \frac{(rF)^2}{24} N_{\text{modes}} \mathcal{N}_{\text{GE}}, \end{aligned}$$

$$\begin{aligned} \mathcal{N}_{\text{GE}}^{I-\text{Coll.}} &= \frac{81}{256} \frac{1}{N_{\text{modes}}} \int \frac{d\Omega_{\mathbf{k}_I}}{(2\pi)^3} k_{\min}^3 P_{\zeta}^2(k_{\min}) \int_{k_{\min}}^{k_{\max}} \left[\frac{d^3 \mathbf{K}_I^1 d^3 \mathbf{K}_I^3}{(2\pi)^6} \right] \sin^4(\theta_I^1) \sin^4(\theta_I^3) \cos^2[2(\varphi_I^1 - \varphi_I^3)] \\ &= \frac{3}{200} \frac{k_{\max}^3}{k_{\min}^3} A_s^2, \end{aligned}$$